

Computer Networks Question Bank

UNIT 1: OVERVIEW OF OSI AND TCP/IP MODEL

Q1. Which layer of the OSI model is responsible for logical addressing?

- A. Data Link Layer
- B. Network Layer
- C. Session Layer
- D. Transport Layer

Q2. In the TCP/IP model, HTTP works at which layer?

- A. Transport Layer
- B. Internet Layer
- C. Application Layer
- D. Network Access Layer

Q3. Which protocol is connection-oriented?

- A. UDP
- B. IP
- C. TCP
- D. ARP

Q4. How many layers are present in the OSI model?

- A. 5
- B. 6
- C. 7
- D. 8

Q5. Which device primarily operates at the Data Link Layer?

- A. Router
- B. Hub
- C. Switch
- D. Gateway

Q6. The process of converting data into signals is handled by:

- A. Application Layer
- B. Physical Layer
- C. Network Layer

D. Session Layer

- Q7. Explain the seven layers of the OSI model with suitable examples.
- Q8. Differentiate between the OSI model and TCP/IP model.
- Q9. Describe encapsulation and decapsulation in networking.
- Q10. Explain the role of protocols in computer networks.
- Q11. Discuss the services provided by the Transport Layer.
- Q12. Explain peer-to-peer communication in layered architecture.
- Q13. Describe the functions of Session and Presentation layers.
- Q14. A file of size 20 MB is transferred over a network with bandwidth 10 Mbps. Calculate the transmission time.
- Q15. A packet of 1200 bytes travels through 4 routers. If each router introduces a delay of 2 ms, calculate the total router delay.
- Q16. A signal propagates through a channel of length 3000 km at a speed of 2×10^8 m/s. Calculate propagation delay.
- Q17. If a message of 5000 bytes is divided into packets of 1000 bytes each, how many packets are generated?
- Q18. Calculate the efficiency if useful data is 900 bytes and total frame size is 1000 bytes.
- Q19. A network has a bandwidth of 100 Mbps and RTT of 20 ms. Calculate bandwidth-delay product.
- Q20. Compare circuit switching and packet switching with suitable diagrams.

UNIT 2: DATA LINK LAYER

- Q21. The LLC sublayer is responsible for?
- A. Routing

- B. Logical Link Control
- C. Error Detection
- D. IP Addressing

Q22. Which addressing method is used at the Data Link Layer?

- A. IP Address
- B. Port Address
- C. MAC Address
- D. Logical Address

Q23. Which protocol is used for flow control?

- A. Stop-and-Wait
- B. HTTP
- C. DNS
- D. ICMP

Q24. Which device works using MAC addresses?

- A. Router
- B. Switch
- C. Repeater
- D. Gateway

Q25. What is the size of a MAC address?

- A. 16 bits
- B. 32 bits
- C. 48 bits
- D. 64 bits

Q26. Which access method is used in Ethernet?

- A. Token Passing
- B. CSMA/CD
- C. FDMA
- D. TDMA

Q27. Explain framing techniques used in the Data Link Layer.

Q28. Explain VLAN concept along with its advantages and disadvantages.

Q29. Explain Stop-and-Wait ARQ protocol with example.

- Q30. Differentiate between Go-Back-N and Selective Repeat protocols.
- Q31. Discuss the working of CSMA/CD.
- Q32. Explain the concept of flow control.
- Q33. Describe HDLC protocol and its frame format.
- Q34. If the propagation delay is 20 microseconds and transmission delay is 10 microseconds, determine the total delay.
- Q35. An Ethernet LAN transmits data at 1 Gbps. Calculate the amount of data transmitted in 2 seconds.
- Q36. For a Go-Back-N protocol with window size 4 and sequence numbers 0–7, determine the sender window after transmitting frames 0–3.
- Q37. A frame of 1500 bytes is transmitted over a 5 Mbps link. Calculate transmission time.
- Q38. A slotted ALOHA system has 40% successful transmission probability. Calculate channel efficiency.
- Q39. Two stations in a CSMA/CD network begin transmission simultaneously. If propagation delay between them is 15 microseconds, determine the time after which collision is detected.
- Q40. Explain token passing mechanism with a suitable example.
- Q41. A channel has $RTT = 100$ ms, Bandwidth = 10 Mbps and Frame size = 10,000 bits. Find the minimum window size required to achieve 100% utilization.
- Q42. A sender transmits frames 0 to 4 continuously using Go-Back-N. Frame 2 is lost while others arrive correctly. Show the sequence of acknowledgements generated by the receiver.
- Q43. A sliding window protocol uses 3-bit sequence numbering and sender window size 5. Determine whether: 1. Go-Back-N is possible 2. Selective Repeat is possible.

UNIT 3: NETWORK LAYER

- Q44. Which protocol is used for logical addressing?

- A. Ethernet
- B. IP
- C. TCP
- D. FTP

Q45. Which device operates at the Network Layer?

- A. Switch
- B. Hub
- C. Router
- D. Bridge

Q46. Which protocol is used to report network errors?

- A. ICMP
- B. ARP
- C. FTP
- D. SMTP

Q47. IPv4 address consists of:

- A. 16 bits
- B. 32 bits
- C. 48 bits
- D. 64 bits

Q48. Which routing protocol is based on distance vector algorithm?

- A. OSPF
- B. RIP
- C. BGP
- D. MPLS

Q49. Which class of IPv4 addresses supports multicasting?

- A. Class A
- B. Class B
- C. Class C
- D. Class D

Q50. Explain IPv4 addressing and subnetting.

Q51. Describe the functions of the Network Layer.

Q52. Differentiate between static and dynamic routing.

Q53. Explain distance vector and link state routing algorithms.

Q54. Describe fragmentation in IPv4.

Q55. Explain IPv6 features and advantages over IPv4.

Q56. Discuss the working of ARP and RARP protocols.

Q57. Determine the network ID and host ID of IP address 192.168.10.25/24.

Q58. An organization needs 50 hosts per subnet. Determine the suitable subnet mask for Class C network.

Q59. Calculate the total number of hosts possible in a /26 subnet.

Q60. Find the broadcast address of network 172.16.5.0/28.

Q61. A router receives a packet with TTL = 1. What happens to the packet?

Q62. Convert the IPv4 address 192.168.1.10 into binary format.

Q63. A packet of size 4000 bytes passes through a network with MTU 1500 bytes. Determine the number of fragments created.

Q64. Given the following network: A to B =1, B to C =2, C to D =3 and A to D= 7. Using distance vector routing, determine shortest path from A to D.

Q65. Differentiate between Distance Vector and Link State Routing.

Q66. Which routing protocol is faster in convergence?

- A. Distance Vector
- B. Link State
- C. Static Routing
- D. Default Routing

Q67. Which technique helps reduce routing loops in Distance Vector Routing?

- A. Flooding
- B. Split Horizon
- C. Fragmentation

D. Multiplexing

Q68. In distance vector routing, routing tables are updated:

- A. Periodically
- B. Only during startup
- C. Once per session
- D. Manually only

Q69. In RIP, if the hop count exceeds 15, what action is taken? Explain with example.

UNIT 4: TRANSPORT LAYER

Q70. Which protocol provides reliable communication?

- A. UDP
- B. IP
- C. TCP
- D. ICMP

Q71. Which field in TCP is used for flow control?

- A. Window Size
- B. Sequence Number
- C. ACK Number
- D. Port Number

Q72. Which protocol is preferred for video streaming?

- A. TCP
- B. UDP
- C. FTP
- D. SMTP

Q73. Which process identifies applications in Transport Layer?

- A. Routing
- B. Fragmentation
- C. Multiplexing
- D. Addressing

Q74. Which mechanism is used by TCP to establish connection?

- A. Two-way handshake
- B. Three-way handshake
- C. Four-way handshake
- D. Sliding Window

Q75. Port number 80 is associated with:

- A. FTP
- B. HTTP
- C. SMTP
- D. DNS

Q76. Explain the services provided by the Transport Layer.

Q77. Differentiate between TCP and UDP.

Q78. Describe TCP three-way handshake with diagram.

Q79. Explain congestion control techniques in TCP.

Q80. Discuss multiplexing and demultiplexing.

Q81. Explain sliding window protocol.

Q82. Describe socket programming concept.

Q83. If the sender window size is 5 and frame size is 1000 bytes, determine maximum outstanding data.

Q84. A TCP segment has sequence number 5000 and carries 1200 bytes. What is the acknowledgment number?

Q85. Calculate throughput if 5 MB data is transferred in 10 seconds.

Q86. If RTT is 50 ms and window size is 10 KB, calculate maximum throughput.

Q87. A TCP connection experiences packet loss after transmission of 16 packets. Explain TCP reaction using congestion avoidance.

Q88. A UDP datagram has total length 1200 bytes and header length 8 bytes. Find payload size.

Q89. Compare connection-oriented and connectionless services with examples.

UNIT 5: APPLICATION LAYER

Q90. Which protocol is used for email transmission?

- A. HTTP
- B. SMTP
- C. FTP
- D. SNMP

Q91. DNS is used for:

- A. File Transfer
- B. Domain Name Resolution
- C. Remote Login
- D. Network Management

Q92. Which protocol is used for secure web communication?

- A. HTTP
- B. FTP
- C. HTTPS
- D. TELNET

Q93. Which protocol is used for remote login?

- A. SMTP
- B. FTP
- C. TELNET
- D. POP3

Q94. Which application layer protocol is used for file transfer?

- A. FTP
- B. DNS
- C. ARP
- D. DHCP

Q95. Which protocol dynamically assigns IP addresses?

- A. HTTP
- B. FTP
- C. DHCP
- D. SMTP

Q96. Explain the working of DNS.

Q97. Describe HTTP request and response methods.

Q98. Differentiate between FTP and TFTP.

Q99. Explain email architecture using SMTP and POP3.

Q100. Discuss the role of DHCP in networks.

Q101. Explain client-server architecture.

Q102. Describe the working of HTTPS and SSL.

Q103. Explain the role of Application Layer protocols in modern Internet services.